

RhythmBlink: Prompting Eye Blinks in Smart Glasses to Prevent Dry Eye Syndrome

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Abstract

Discreet, lightweight, optical see-through smart glasses are an emerging device form factor with high potential for integration into our daily lives. However, prolonged use of their displays can reduce blink rates, leading to discomfort and, eventually, Dry Eye Syndrome (DES). Previous studies have proposed methods to induce blinks in VR by triggering a reflex response using either visual or physical stimuli. While reported to be effective, such techniques may not be applicable to see-through smart glasses as such induced blinks may result in situational impairments. Accordingly, we propose RhythmBlink, a gamified system designed to prompt voluntary blinking. Our study shows that RhythmBlink increases blink rates (16.18 blinks/minute) compared to the baseline (10.48 blinks/minute), demonstrating its effectiveness in encouraging blinks. While it led to increased cognitive workload, the result suggests that gamification can improve eye health by increasing blink rates during smart glass use.

CCS Concepts

• Human-centered computing → Interaction design.

Keywords

Smart Glasses, Dry Eye, Blink Rate

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1 Introduction

The development of smart and Augmented Reality (AR) glasses targeting everyday use scenarios is advancing rapidly. Recent releases, such as the XREAL One Pro [34], combine high Field of View (FoV) displays (e.g., 57 degrees), stable visual rendering, and discreet form factors; they are designed for prolonged use by diverse end-users. Indeed, such smart glasses can be valuable tools to efficiently perform daily tasks, such as receiving notifications

or interacting with navigation systems. Compared to more bulky Virtual Reality (VR) headsets that are impractical for outdoor use due to their size and weight, smart glasses are designed to support a range of mundane applications. Furthermore, as these devices become more prevalent and integrated into daily routines, their use for visually demanding tasks, such as reading documents or participating in virtual meetings, is expected to increase.

However, visually demanding tasks can reduce blink frequency, which in turn can lead to discomfort and dry eye syndrome (DES) [27]. The symptoms of DES include eye irritation, heightened sensitivity to light, and, in severe cases, blurred vision [20]. DES occurs frequently during prolonged use of traditional digital devices such as desktop computers, laptops, and mobile devices and also with various forms of Optical Head-Mounted Displays (OHMD) such as both virtual and augmented reality headsets [36]. Indeed, end-user optical see-through smart glasses typically have a reduced distance between the eye and the display, a factor that can significantly elevate the occurrence of DES [28]. There are several methods to alleviate DES. Preventative strategies, such as viewing content at an appropriate visual angle, taking regular breaks, and performing eye exercises [14], are commonly suggested but can be difficult to sustain due to the explicit, long-term effort required. Eye drops with artificial tears are a widely used remedy [27] that addresses the symptoms but not the causes of DES. Finally, one of the frequently recommended treatments for DES is to blink more often. Based on the prevalence of this condition, and its emerging association with prolonged smart glass use, we believe it necessary to develop methods to support appropriate blinking rates during OHMD use.

Prior work has also examined this issue and, indeed, several methods to induce blinking have been proposed [4, 6, 22, 37]. These methods typically rely on inducing reflexes to trigger blinks, for example, by presenting a flashing light or rapidly approaching object. They have also typically been developed for desktop computers [6] or VR headsets [37]. However, we believe this fundamental approach, or relying on induced reflex reactions, is inappropriate for smart glasses designed for real world settings, as involuntary blinks may lead to situational impairments [18, 32] at inappropriate moments. Essentially, the real-world tasks users may opt to complete while wearing their smart glasses may include a wide range of safety-critical tasks (e.g., food preparation, driving, manual work). Systems that automatically activate the blink reflex may lead to additional hazards and should be avoided. Furthermore, the reduced size of smart glass displays (e.g., 50 degrees) compared to VR HMDs (e.g., up to 100 degrees) may require new approaches: visual cues that work in an immersive setting may be ineffective on smart glasses.

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Accordingly, this paper introduces RhythmBlink, a gamified system that encourages voluntary eye blinking while minimizing interference with a user's main task performance when wearing smart glasses. In the design of this technique, we are particularly interested in primary tasks that involve intensive visual activity, such as visual inspection or detailed manual work. RhythmBlink involves an explicit, predictable visual cue designed to gently remind users to blink. We conduct a study evaluating its performance against a baseline (no intervention) and a best-in-class approaching object effect from prior work [37] while users complete a simple search-based puzzle task that requires intensive visual activity. We evaluate its effect by recording blink rate and subjective workload. The results indicate that participants blinked an average of 16.18 times per minute in RhythmBlink task, a significant improvement over the baseline condition (10.48 times per minute). However, the study also found an increase in workload, suggesting a potential trade-off between prompting blinks and elevated mental effort. While task performance differences were not statistically significant, the potential impact of increased workload warrants further investigation. We conclude that graphical aids to remind users to blink can be effective and may help reduce eye strain during prolonged smart glass use.

2 Related Work

2.1 Eye Blinking Pattern and DES

Human blinking patterns play a crucial role in maintaining eye health and are closely linked to the causes of DES. Eye blinking can be categorized into voluntary blinks, which are consciously controlled, and involuntary blinks, which occur without conscious effort [30]. Involuntary blinks include both spontaneous and reflex blinks. In the normal state, the spontaneous eye blink rate (SEBR) typically ranges from 15-20 blinks per minute. However, it can drop significantly during visually demanding tasks, such as prolonged screen use or focused activities, to as low as 5-6 blinks per minute [9, 24]. A reduced blink rate results in faster tear evaporation, which can cause symptoms of DES, including itching, burning, light sensitivity, and blurred vision [20, 27].

2.2 Stimulating Eye Blinks and Situational Impairment

To overcome DES, various methods have been explored to stimulate eye blinks. Software techniques such as blurring, flashing [4], and playing animations [22] have proven effective when applied to desktop monitors. Additionally, an approaching object has been shown to induce the blink reflex in a VR environment [37]. Hardware solutions also exist, such as Wink glasses, which fog the lenses if the user does not blink for five seconds [7], and DualBlink [6] which added cues such as LED flashes, physical taps, or air puffs to glasses. Utsumi et al. [29] proposed glasses with micro-blowers and inertial sensors to detect blink timing and induce blinking. While diverse and interesting, these methods are unsuitable for daily use with smart glasses, as they can cause situational impairments [18, 32]. In particular, previous blink-inducing methods relying on reflex responses to visual stimuli such as flashes or approaching objects may lead to situational impairments [4, 37]. Such impairments are particularly problematic for smart glasses as these devices may

be used in mobile [26] contexts or while performing potentially hazardous tasks (e.g., food preparation, driving). This suggests the need for a new approach that prompts voluntary blinks, rather than relying on involuntary reflex responses, to minimize the situation impairments that may result.

2.3 Secondary Task and Gamification

To design cues to support voluntary blinking, we leveraged the literature on secondary task assessment procedures. In studies deploying such methods, a secondary task is performed simultaneously with a primary task of interest, and performance on the secondary task is used as a measure to assess cognitive workload and task prioritization [21]. Key factors in designing a secondary task include its sensitivity to primary task demands, participant engagement, and ease of implementation [10]. Secondary tasks should also be discretionary, meaning they should not interfere with primary task performance, even if participants ignore them. Additionally, instructions in such studies emphasize that the accuracy of the primary task takes precedence over the secondary task. Secondary task methods are widely used in studies of workload and have also been applied to smart glasses [2] to assess performance in tasks such as navigation. Inspired by this approach, we sought to design RhythmBlink following the principles outlined for creating effective secondary tasks.

To increase engagement with our design, we were also inspired by the literature on gamification. This approach seeks to enhance engagement by incorporating game design elements into non-game contexts [33]. Elements like scores, dashboards, and achievements/badges motivate users to actively participate to complete tasks [5, 11, 15]. Additionally, by appealing to basic psychological needs for autonomy and competence, gamification can enhance motivation [1]. However, overly competitive designs can demotivate users [11]. Gamification approaches have also been previously applied to eye movement patterns, such as blinking, demonstrating the potential to incorporate such physiological actions into game play [23, 25, 31]. We believe this design approach opens new possibilities for engaging users in novel experiences.

3 Method

3.1 Study Design

We conducted a study to assess the performance of different techniques to promote eye-blinking behavior while wearing smart glasses and performing a visually intensive primary task. The study was approved by the local IRB.

3.1.1 Primary Task. We selected a puzzle for the visually intensive primary task: a word search. This task requires participants to find a set of target words in a 12x12 grid of letters. The role of this task in the study is to show how the different secondary tasks impact the participant's performance in the main task. To ensure no biases in the puzzles presented, we generated a large set of 200 random word search puzzles, standardizing the difficulty level across each. Specifically, all puzzles featured a fixed number of word lengths of four different lengths (two four-letter words, three five- and six-letter words, and two eight-letter words), all selected randomly from a dictionary of the most common ten thousand



Figure 1: Image of the equipment use in the experiment

English words [8]. We also randomized and equalized the orientation of the words among different directions (three horizontal and vertical, two diagonal right and left), avoiding reversed words. In addition, the distribution of letters in the non-word spaces followed the relative frequency of each character in English. With this set, each participant received one or more unique but equally difficult puzzles in each task.

3.1.2 Secondary Task. We present three secondary tasks in this study. One is a *baseline*, in which no cues were presented. The other two methods were designed to prompt eye blinks. The first was the approaching object effect proposed in prior work [37]; we follow the original implementation closely. It features a white circle that expands from a radius of 25px to 500px over 300ms as it moves toward the bottom-left corner of the display. In the remainder of this paper, we refer to this cue as the *reflex* task. The second was *RhythmBlink*, a gamified animation. In *RhythmBlink*, a green ring (of 105px diameter and 20px thick) appears at the center of the screen each time a blink is to be cued. At the same time, a gray circle expands from a radius of 0 to 150px. While the outer edge of the gray circle intersects the ring (850-1050ms) it turns blue. Participants score a point by blinking when this color change is presented. Additionally, a short tone indicates the start of each animation, and when a blink is detected, a higher frequency tone is played. In both of the cues, the desired time interval between blinks is set to 4.3 seconds, supporting at least 14 blinks per minute [9, 24]. Thus, the approaching object cue is presented after 4 seconds pass without a blink, while *RhythmBlink* is presented 3.35 seconds after a blink detection.

3.2 Apparatus

To detect eye blinking, participants wore the Pupil Labs Neon [16] with two infrared (IR) eye cameras, each paired with 850nm infrared illuminator LEDs. These cameras are positioned at the nose bridge of the glasses to capture eye videos effectively. The data received by the Pupil Labs Neon was transmitted to a Motorola Edge 40 Pro smartphone, which then forwarded it to a laptop connected to a pair of Xreal Air 2 Ultra [35] smart glasses. We used the built-in Neon blink detection algorithm to detect eye blinks [17]. In addition, the Xreal glasses offer the advantage of easily implementing desired tasks by simply mirroring the laptop screen. The device provides a 52-degree field of view and displays in 1920x1080. A custom 3D-printed nose bridge was created to combine these two devices. This combined unit is shown in Figure 1.

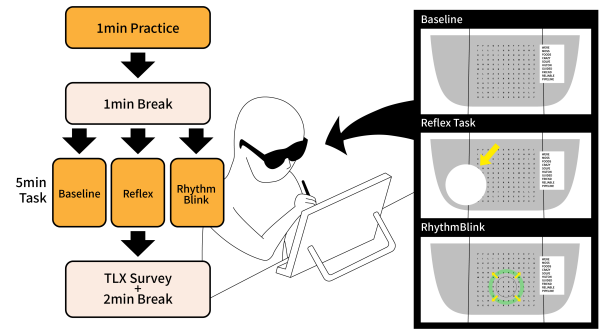


Figure 2: Illustration of experimental procedure

3.3 Participants

Eighteen participants (12 male and 6 female) were recruited from the local campus, with an average age of 24.44 ($SD = 3.79$). While some participants had previous experience with OHMDs such as VR devices, none of them had prior experience with smart glasses. Recruitment focused on individuals without visual impairments, such as color blindness, to ensure they could accurately perceive color-based interactions. In addition, participants were required to have normal vision without the need to wear glasses to ensure they could wear the study equipment comfortably. One participant used contact lenses to achieve normal vision.

3.4 Procedure and Data Analysis

The participants were invited to a quiet room where no one was present except the researcher to minimize distractions. Detailed instructions were provided, explaining that the primary task would involve completing a series of word search puzzles for 5 minutes while wearing smart glasses. After an overview of the experimental protocols, participants who opted to continue the study signed a consent form and completed a demographic survey. They were told the study focused on analyzing eye movement patterns without explicitly mentioning blinking to avoid influencing their behavior; however, they were instructed in the intended use of the *RhythmBlink* visualization. The word search puzzles were printed on A4-sized paper and placed on a book stand with a 50-degree angle to reduce discomfort from the device's weight. Participants were instructed to mark the target words displayed on the right side of the smart glasses screen, a design choice intended to minimize distractions [3]. Since there are six possible sequences in which the three conditions (baseline, reflex, and *RhythmBlink*) can be presented, we applied each sequence to three participants, ensuring balanced exposure across conditions. This resulted in a total of 18 participants divided into six groups. This counterbalancing method ensured the elimination of order effects.

Each task followed a structured sequence: a 1-minute practice session to familiarize participants with the interaction and device, followed by a 1-minute rest period. Then, the blink patterns are collected during the subsequent 5-minute main task. Data collected included blink detection and secondary task animation timings. After each task, participants completed the NASA-TLX survey after

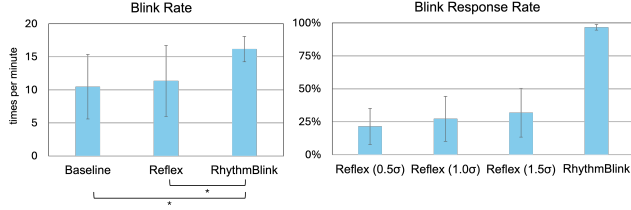


Figure 3: Bar plots of average blink rate per minute (left) and average blink response rate (right) for each task. Error bars show standard deviation and brackets indicate statistically significant differences ($p' < 0.05$).

each condition [12] that assessed six items: mental demand, physical demand, temporal demand, performance, effort, and frustration. Each item was rated on a 0 to 20 with no weights [13]. Then, we provided a 2-minute break to mitigate fatigue before proceeding. Each condition lasted approximately 10 minutes, including task execution, surveys, and rest periods. Therefore, the total duration of the experiment was approximately 30 minutes. Figure 2 provides an overview of the experimental procedure. To emphasize the importance of the primary task, participants were informed that the top three performers in the word search would receive an additional 7\$ reward, supplementing their base compensation of 7\$.

4 Result

The total number of blinks across all tasks and participants was 3607. Before analyzing the results, we filtered out "double blinks." These occurred if the time interval between two consecutive blinks was shorter than a specified threshold (in this case, 500ms) and were considered as a single blink. We found and culled 185 (5.13%) double blinks leaving a final set of 3422 blinks. We then conducted a repeated measures ANOVA with Bonferroni-corrected post-hoc tests to identify significant differences in performance across the three interaction tasks. We use p' to indicate Bonferroni-corrected p-values throughout this paper.

4.1 Blink Rate

Figure 3 (left) shows the average blink rate per minute for each condition. A repeated measures ANOVA revealed a significant difference in blink rates between tasks ($F = 12.8$, $df = 2$, $p < .001$). RhythmBlink had the highest blink rate ($M = 16.18$, $SD = 1.93$) compared to the other tasks. There was no significant difference between the baseline ($M = 10.48$, $SD = 4.88$) and reflex task ($M = 11.36$, $SD = 5.36$) ($p' = 1.000$). However, RhythmBlink significantly differed from both the baseline ($p' < .001$) and the reflex task ($p' = 0.002$).

4.2 Blink Response Rate

In previous work, the average response time for the approaching ball effect was 0.67 seconds, with a standard deviation of 0.58 seconds [37]. Based on this, we defined three thresholds by adding 0.5, 1.0, and 1.5 times the standard deviation to the mean, resulting in thresholds of 0.96 seconds, 1.25 seconds, and 1.54 seconds, respectively. Figure 3 (right) illustrates the average response rates for

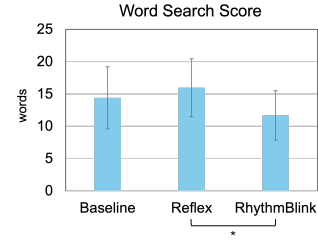


Figure 4: Bar plots of average word search score in each condition. Error bars show standard deviation and brackets indicate statistically significant differences ($p' < 0.05$).

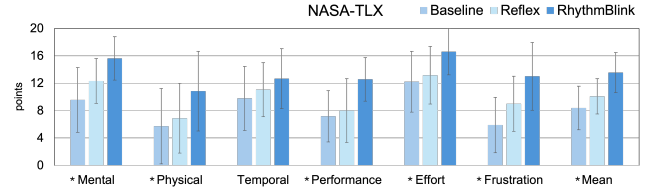


Figure 5: Bar plots of average NASA-TLX score for each question and average of total score. The error bar means standard deviation. The asterisk(*) next to the label indicates statistically significant differences ($p < 0.05$).

each threshold across tasks. For the reflex task, the response rate remained below 32% across all thresholds, with averages of 21.56% ($SD = 13.70\%$) for the 0.96-second threshold, 27.31% ($SD = 17.12\%$) for the 1.25-second threshold, and 31.98% ($SD = 18.54\%$) for the 1.54-second threshold. In contrast, RhythmBlink demonstrated a substantially higher response rate, achieving an average of 96.68% ($SD = 2.12\%$).

4.3 Word Search Performance

Figure 4 shows the word search scores for each condition. A repeated measures ANOVA test indicated a difference in scores across tasks ($F = 7.93$, $df = 2$, $p = 0.001$). However, a pairwise post-hoc test showed no significant difference between the baseline ($M = 14.39$, $SD = 4.80$) and reflex task ($M = 15.94$, $SD = 4.48$) ($p' = 0.642$). RhythmBlink ($M = 11.67$, $SD = 3.80$) differed significantly from the reflex task ($p' = 0.002$), and showed a marginal difference from the baseline ($p' = 0.053$). While this result does not meet the conventional threshold for statistical significance, the small p-value suggests a potential effect that may become significant with a larger sample size.

4.4 Workload

Figure 5 indicates that NASA-TLX scores for all metrics were lowest in the baseline task, followed by the reflex task, and highest in the RhythmBlink. A repeated measures ANOVA revealed no significant difference in temporal demand ($p = 0.079$). However, mental demand ($p = 0.011$) and all other metrics ($p < .001$) showed differences. For the overall workload scores (rightmost area in the

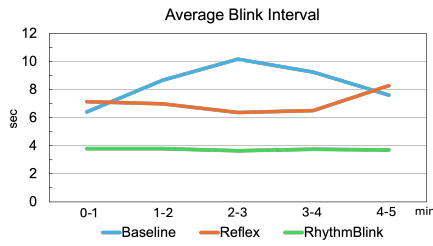


Figure 6: Line plots of average intervals between detected blink timestamps over experiment duration of each task. The x-axis represents time intervals in 1-minute.

graph), baseline scored 8.39 ($SD = 3.19$, $min = 3.83$, $max = 13.67$), reflex scored 10.07 ($SD = 2.57$, $min = 6.50$, $max = 14.33$), and RhythmBlink scored 13.55 ($SD = 2.92$, $min = 6.83$, $max = 17.67$).

5 Discussion

5.1 Result Analysis

RhythmBlink led to significantly elevated blink rates compared to both baseline ($p' < .001$) and reflex tasks ($p' = 0.002$). The average blink rate of RhythmBlink was 16.18 times per minute, modestly exceeding the recommended rate of 15 and demonstrating its potential to address DES [9, 24]. Furthermore, as shown in Figure 6, RhythmBlink maintained a consistent blink interval throughout the study, indicating its ability to sustain a steady blinking pattern over time. However, this improvement came with trade-offs as NASA-TLX scores were higher than in the baseline and reflex conditions, indicating an increased workload for the primary task. Although word search performance decreased, the difference was not statistically significant ($p' > 0.05$) compared to the baseline.

In comparison, the reflex task showed limited efficacy. Although the average blink rate increased over baseline, the change was not significant ($p' = 1.000$). The blink response rate was notably low at less than 32%, suggesting that the approaching ball effect was insufficient to induce a reflex response in the smart glasses we used. This result stands in contrast to prior work on a VR headset that led to a high blink rate [37]. As shown in Figure 6, the baseline condition initially exhibited an increase in the blink interval, followed by an expected decrease as eye fatigue accumulated over time [19]. In contrast, the reflex task demonstrated shorter blink intervals during the initial stages, which may be due to reflex-induced blinking. However, after 4 minutes, the blink interval began to increase, suggesting that participants had adapted to the stimulus, reducing its effectiveness. We conclude that the reflex task did not yield meaningful improvements in blink behavior while contributing to an elevated workload.

5.2 Limitation and Future Work

One of the limitations of this study lies in the design of the primary task. Although the word search puzzle was intended to reduce blink rates and thus lead to DES, it did not reduce blinking as much as expected ($M = 10.48$ in the baseline). Future work could explore alternative tasks to achieve a more pronounced reduction in baseline blink rates. Another limitation is the short duration of

the experiment. The 5-minute tasks may not have allowed participants enough time to adapt, making it difficult to fully observe the effects of each intervention. Extending the study duration could provide more reliable insights. Moreover, while efforts were made to standardize the difficulty of the word search puzzles, individual differences in task performance were evident. Across tasks, participants found an average of 14 words per condition, with a relatively high standard deviation of 3.48. This high variability may be the result of some puzzles being more challenging than others. Increasing the number of trials for each task would help account for differences in word search difficulty, minimizing the impact of this factor and enhancing the reliability of the findings.

RhythmBlink interface's placement in the central field of view may have contributed to increased workload, as indicated by higher NASA-TLX scores. Future iterations should consider leveraging peripheral vision to minimize visual interference with the primary task [2]. This adjustment could reduce workload while maintaining the effectiveness of blink stimulation. Finally, while RhythmBlink successfully increased the blink rate beyond the recommended threshold of 15 blinks per minute, its animation duration could be optimized to balance blink stimulation and workload. Fine-tuning the animation timing and frequency could help achieve the desired outcomes with reduced disturbance.

6 Conclusion

We introduced RhythmBlink, a gamified secondary task designed to address the issue of decreased blink rates and the resulting DES that occurs when users perform visually demanding tasks with smart glasses. The primary task involved a word search puzzle, which required significant visual focus and naturally reduced blink rates while providing a quantifiable measure of task performance based on the number of words found. Compared to the reflex task, RhythmBlink effectively increased blink rates to an optimal level and maintained consistent time intervals throughout the experimental duration. However, while primary task performance did not significantly decline, participants did report increased workload, indicating that the user experience was not entirely satisfactory. These limitations suggest opportunities for improvement, such as leveraging peripheral vision to minimize visual interference and enhance usability. Future work should explore these ideas to create smart glass systems that go easier on the eyes.

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